

Peter Eremeev

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EDUCATION

London School of Economics Sep 2026 – Jun 2027
MSc Econometrics and Mathematical Economics (incoming)

University of California, San Diego Sep 2024 – Jun 2026
B.S. Joint Mathematics–Economics with High Distinction
Mathematics Honors Program

RESEARCH INTERESTS

time series econometrics, nonparametric methods, macroeconomics

WORKING PAPERS

Nonparametric Spectral Tests for Stationarity in Nonlinear Time Series 2026

Honors thesis, advised by Danna Zhang.

A nonparametric test for stationarity based on the spectral density, accommodating nonlinear, non-Gaussian data-generating processes and retaining power against smoothly time-varying alternatives. Establishes a joint CLT for the L^2 -distance estimator under GMC dependence and applies the resulting test to credit-spread data where piecewise-stationary alternatives fail.

RESEARCH EXPERIENCE

Independent Study Jun – Sept 2025

Supervisor: Prof. Danna Zhang — Time series analysis

Stationarity and linear processes; ARMA modeling and forecasting; estimation, diagnostics, and model selection. Informal introduction to spectral analysis.

Research Assistant, The Economic Research Lab, UC San Diego Mar – Jun 2025

Supervisor: Maddison Erbabian (PhD candidate, Economics) — Event study on educational spillovers from universities.

Built Python pipeline for automated geo-coordinate scraping via the Google Maps API. Constructed location database with crosswalks to household microdata. Verified and modified replication packages; ported R GIS objects into Python.

TEACHING EXPERIENCE

Academic Tutor, College of San Mateo 2022 – 2024

Tutored introductory microeconomics and statistics.

HONORS AND AWARDS

Provost Honors, UC San Diego 2024 – 2026

Paper accepted, Stanford Undergraduate Honors Symposium 2023

COURSEWORK

Mathematics: Real Analysis, Probability, Stochastic Processes, Mathematical Statistics, Linear Algebra

Economics: Intermediate Microeconomics, Short-run Macroeconomics (Honors), Long-run Macroeconomics, Econometrics

SKILLS AND LANGUAGES

Programming: R, Python, Stata (basic), Git, $\LaTeX$

Languages: English, Russian

REFERENCES

Danna Zhang

Associate Professor of Mathematics

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Nir Jaimovich

Professor of Economics

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